Midterm Exam

Course Title	Calculus 2	Course Code	3020102
Semester	Second	Academic Year	2025/2026
Exam Date	14/4/2026	Day	Tuesday	Time: 2:30-3:20

Student Name		Student ID No.		Section	

GENERAL INSTRUCTIONS

1. Students are not allowed to leave the exam room during the first 30 minutes of the exam..
2. Late students are not permitted to complete their exams after the first 30 minutes unless authorized by the proctor (no students left the exam session), knowing that no additional time will be granted.
3. Students must use a blue or black pen. Pencils are not allowed.
4. Consumption of any food or beverage is prohibited except in medical cases.
5. All electronic devices (e.g., cell phones) are prohibited.
6. All cell phones must be turned off and placed on the invigilator's desk in the exam room.
7. All notes and reading materials must be placed on the invigilator's desk in the exam room.
8. Students are not permitted to speak to or ask questions of their classmates. If a student has a question, he or she must raise his or her hand to be assisted by the proctor.
9. Write the full information required on this page before commencing your exam.
10. Students are not allowed to leave the room once the exam commences for any reason unless it is absolutely necessary.
11. Reported student cheating cases, misbehavior, or other student misconduct on examination sessions will be reported to the Investigations Committee as per internal bylaws.

I have understood and abide with the exam policy. **Student's signature:** _____

Question No.	Question Type	CILOs assessed	Marks Weight	Scored Marks	Marker signature	Student Score
1-6	Multiple choice	CLIO 1,2,3,4,5,6	12 (40%)			
7-9	Easy		18 (60%)			
Total			30 (100%)			

Moderation Review			
Comments			
Marks Reviewed by		Signature & Date	

Q(1)	Q(2)	Q(3)	Q(4)	Q(5)	Q(6)

Q(1) $csch(lnx) =$

a) $\frac{2x}{x^2+1}$ b) $\frac{x}{x^2-1}$ c) $\frac{2x}{x^2-1}$ d) $\frac{x}{x^2+1}$

Q(2) The volume of the solid generated when the region between the graphs of the equations $y = \sqrt{2x}$, $y = \sqrt{x+3}$ and $x = 0$, is revolved about the x −axis, is:

a) $\frac{\pi}{2}$ b) 2π c) $\frac{9\pi}{2}$ d) $\frac{\pi}{4}$

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Q(3) The arc length of the curve $y = \cosh x$, from $x = 0$ to $x = e$, is:

a) $\sinh(e)$ b) 1 c) $\sinh(2e)$ d) $\cosh(e)$

Q(4) $\lim_{x\to\infty} xe^{-x}\ln(x+3) =$

a) 0 b) ∞ c) 3 d) e^3

Q(5) $\int_0^2 \frac{1}{\cosh x - \sinh x}\,dx =$

a) $1 - e^2$ b) $e^2 - 1$ c) $1 - e^{-2}$ d) e^2

Q(6) $\int_0^{\frac{\pi}{2}} \cos^3(x)\sin^2(x)\,dx =$

a) $\frac{1}{15}$ b) $\frac{3}{15}$ c) $\frac{8}{15}$ d) $\frac{2}{15}$

(6 marks)

Q(7) Evaluate $\int x\sec^2(x)\,dx$.

u	dv	
x	$\sec^2 x$	+
1	$\tan x$	−
0	$-\ln\lvert\cos x\rvert$	

$$\int x\sec^2 x\,dx = x\tan x + \ln|\cos x| + c$$

Q(8) Evaluate $\int \sin x(\sin(2x) + \cos(3x))dx$. (6 marks)

$$\int \sin 2x \sin x + \sin x \cos 3x \, dx$$

$$\int \sin 2x \sin x \, dx + \int \sin x \cos 3x \, dx$$

$$\frac{1}{2}\int \cos x - \cos 3x \, dx + \frac{1}{2}\int \sin(4x) + \sin 2x \, dx$$

$$\frac{1}{2}\left(\sin x - \frac{\sin 3x}{3}\right) + \frac{1}{2}\left(-\frac{\cos 4x}{4} - \frac{\cos 2x}{2}\right) + C$$

$$\frac{\sin x}{2} - \frac{\sin 3x}{3} - \frac{\cos 4x}{4} - \frac{\cos 2x}{2} + C$$

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Q(9) Use trigonometric substitution to show that: prove that (6 marks)

$$\int (1+\ln x)\sqrt{1+x^2(\ln x)^2}\,dx = \int sec^3(\theta)\,d\theta.$$

$$\int (1+\ln x)\sqrt{1+(x\ln x)^2}\,dx$$

$$x\ln x = \tan\theta \Rightarrow (1+\ln x)\frac{dx}{d\theta} = \sec^2\theta$$

$$dx = \frac{\sec^2\theta\,d\theta}{(1+\ln\theta)}$$

$$\int \cancel{(1+\ln x)}\sqrt{1+\tan^2\theta}\,\frac{d\theta(\sec^2\theta)}{\cancel{(1+\ln x)}}$$

$$\int \sqrt{\sec^2\theta}\,(\sec^2\theta)\,d\theta$$

$$\int \sec\theta\,(\sec^2\theta)\,d\theta$$

$$\int \sec^3\theta\,d\theta$$ ✱

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✱ لجنة سنابل البوليتكنك

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